How Many Independent Models Does an Open-Model Ecosystem Contain? Evidence at the Exact Conditional Null

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Abstract

Whether independently developed language models fail on the same inputs bears directly on systemic risk, ensemble reliability, and model-as-judge evaluation. Recent work has shown that the answer depends on the null model chosen, and concluded that monoculture inference is therefore *subjective*, leaving open which null is appropriate [Jo et al., 2026]. We argue that one null is canonical rather than arbitrary. Because the row and column margins of a model-by-item outcome matrix are the jointly sufficient statistics of the Rasch family, the uniform distribution over margin-matched matrices is the unique *exact, fit-free* conditional null for that entire family: it estimates no parameters, applies no penalty, and makes no identification choice. We report what the answer is at that null, on 1,228–1,373 open models per benchmark across five benchmarks — a population three times larger than previously analysed.

At the mean level there is nothing to find, and this is a theorem rather than a measurement: mean pairwise co-failure is a function of the item margins alone, so its excess over any margin-preserving null is identically zero. We show this degeneracy has been proved independently in three literatures — as Schluter's V-ratio in ecology, as Kuder–Richardson / Cronbach's α in psychometrics, and as Fleiss's observed agreement for the multi-category case — and give the translation between them, since none is cited in the model-evaluation literature. Empirically the measured excess matches the closed form to 10^{-16} on all five benchmarks. Our pre-registered dispersion statistic came out *below* its null on three of five benchmarks and above on two; its pre-registered kill condition fired and we report it as refuted.

What survives is second-order and large. On ARC, conditioning on the margins leaves residual correlation with root-mean-square 0.2483 against 0.0449 under the exact null — a 5.5-fold excess — and six eigenvalues above the null spectral edge; across the five benchmarks the excess ranges from 2.8× (GSM8K) to 11.8× (HellaSwag). The effect is not an artifact of near-duplicate models: removing every model that agrees with a retained model above 0.95 discards 36–64% of each population and moves these ratios by at most 1.14×. We deliberately do *not* report this as an "effective number of independent models": we show that the participation ratio is algebraically $N/(1 + (N-1)\overline{R_{ij}^2})$, so it cannot distinguish one weak global factor from many tight clusters, and we give the diagnostics that can. Misspecification controls show that two-parameter discrimination heterogeneity cannot generate the observed effect (rms ≤ 0.076), but a low-dimensional multi-ability structure generates much of it (rms 0.162); the observed spectral signature resembles the latter, not duplicated model families. The honest reading is therefore that open-model failure is governed by a handful of latent dimensions beyond unidimensional ability. We also reconcile two apparently contradictory statistics exactly — the excess variance is 29× the null while its covariance with the margin-model prediction is strongly negative — and show the effect is not an artifact of duplicate models, chain mixing, or the choice of spectral functional.

We also state what the test cannot see, because it is not a detail. A planted-signal power study gives full power against shared failure modes at a planted rms of 0.048, well below the 0.099–0.294 observed, and detects copying of a reference model from a copy rate of

0.1; but against the alternative in which every model shares one item-difficulty profile and differs only in overall ability — arguably the strongest form of monoculture — power equals the nominal α of 0.05. That alternative *is* the null, so the test has no power against it by construction. This is the Narcissus effect of Colwell & Winkler [1984] in exact form: we measure departures from a shared difficulty profile and cannot measure the shared profile itself. A pre-registered test on real model responses (not just correctness) finds that our own critique does *not* rescue one specific published claim: the reported association between model accuracy and error agreement attenuates only 12% under conditioning and remains significant, so that finding is reported as robust rather than reinterpreted.

1 Introduction

If many deployed models share blind spots, diversification is illusory: an ensemble, a panel of model judges, or a market of competing screening vendors supplies far less independent evidence than its headcount suggests. A growing literature measures how often models err together, and almost all of it shares one pattern: record which models fail which items, then compare the rate at which pairs fail together against a baseline in which they are assumed independent.

Jo et al. [2026] showed that this comparison is highly sensitive to the assumed baseline — conditioning on item difficulty leaves residual correlations "substantially attenuated", with "some even flipping from strongly positive to slightly negative" — and proved that a sufficiently expressive null can explain away any observed dependence. Their conclusion is that monoculture is a context-dependent inference problem, and they state explicitly that their framework "does not prescribe which null is right".

We take up that open question. Our claim is that the ladder of nulls has a canonical rung. The row and column margins of a binary outcome matrix are the jointly sufficient statistics of the Rasch model; conditioning on them therefore removes the entire nuisance family without estimating any of it. The resulting null — the uniform distribution over margin-matched matrices, known in ecology as the fixed-fixed null and in psychometrics as the Rasch conditional null — requires no gradient ascent, no regularisation, and no identification convention. It is the point at which "we conditioned on ability and difficulty" stops being an analyst's choice and becomes an exact statement.

Contributions.

1. **A canonical stopping point.** We identify the exact conditional null implied by Rasch sufficiency as the non-subjective answer to the null-selection problem posed by Jo et al. [2026], and report what the answer is there (Section 4).

2. **A translation table across three literatures.** The degeneracy of mean co-failure under margin-preserving nulls is Schluter's V-ratio result in ecology [Schluter, 1984, Gotelli, 2000, Gotelli & Ulrich, 2012], Kuder–Richardson/Cronbach's α in psychometrics [Kuder & Richardson, 1937, Cronbach, 1951], and Fleiss's observed agreement in the multi-category case [Fleiss, 1971]. None is cited in the model-evaluation literature. We state the identities and attribute them (Section 3).

3. **A calibrated measurement of residual structure at scale**, over 1,228–1,373 models per benchmark, together with a negative result about the statistic most tempting to report: the participation ratio is a monotone transform of mean squared residual correlation and does not count independent models (Section 6).

4. **An exact reconciliation of two contradictory-looking results**, and robustness to deduplication, mixing, and functional choice (Sections 6.6–6.7).

5. **A released substrate and estimator**, harvested at zero inference cost, with the harness pitfalls documented (Section 5).

What we do not claim. We do not claim prior measurements were computed incorrectly, nor that language models are in fact diverse. We claim that a class of baseline cannot separate shared behaviour from shared item difficulty; that a canonical baseline exists; and that at that baseline the concentration conclusion survives in a different and more defensible form than the one usually reported.

2 Setup

Let $F \in \{0,1\}^{N\times M}$, $F_{im} = 1$ iff model i fails item m. Write $r_i = \sum_m F_{im}$, $c_m = \sum_i F_{im}$, $p_i = r_i/M$, $f_m = c_m/N$, $\bar{f} = \frac{1}{M}\sum_m f_m$, and $C_{ij} = \frac{1}{M}\sum_m F_{im}F_{jm}$.

3 The mean is degenerate, and this is known

Lemma 1 (Mean co-failure depends only on item margins)**.** $O = \dfrac{1}{MN(N-1)}\sum_m (c_m^2 - c_m)$.

Proof. $\sum_{i\neq j} F_{im}F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$ since $F_{im} \in \{0,1\}$. □

Consequently any randomisation preserving the column margins leaves O invariant with zero variance, and the standardised effect size for O is $0/0$. Note the scope precisely: the invariance follows from preserving the *item* margins alone. A null preserving only the model margins does not leave O invariant, so Lemma 1 does not by itself impugn analyses whose baseline conditions on model accuracy.

O is the double-fault measure. For a machine-learning reader, O is not an unfamiliar quantity: mean pairwise co-failure is exactly the *double-fault* diversity measure, one of the ten catalogued by Kuncheva & Whitaker [2003] in the classifier-ensemble literature. Lemma 1 therefore says that a standard ensemble-diversity measure is determined by the item margins. The effect is visible in the canonical diversity-regression design of that literature: over 5,999 sampled panels drawn from our ARC response data, mean member accuracy and panel size alone explain $R^2 = 0.9928$ of majority-vote accuracy, and the standardised regression coefficients on member accuracy and double-fault diverge to $+316$ and $+275$ — the signature of the near-perfect collinearity Lemma 1 predicts. We deliberately do *not* present this as a novel discovery: Kim [2026], posted days before this analysis, independently audits five diversity-related measures — including double-fault and disagreement — as predictors of realised majority-vote gain over 31,900 subsets of 30 LLMs on MMLU-Pro, and derives the exact algebraic identity strict diversity = disagreement + double -fault together with $1 - \overline{\text{Acc}} =$ double-fault + $\frac{1}{2}$disagreement, showing these three classical statistics are rank-deficient once mean accuracy is controlled for. That analysis uses no margin-preserving null and a different target (gain over the best member rather than raw ensemble accuracy), so it is independent corroboration by different machinery rather than a restatement of Lemma 1; we cite it because a reader evaluating whether diversity metrics predict LLM panel accuracy should have both results in view.

This is not new, in three separate fields. In ecology it is the degeneracy of Schluter's V-ratio [Schluter, 1984]. Gotelli [2000] states plainly that "the V ratio was not used in this analysis because retaining row and column totals as in SIM9 does not ever change the index", and Gotelli & Ulrich [2012] generalise the warning to the Sørensen, Simpson, Morisita and NODF families. In psychometrics the same content appears as reliability:

Lemma 2 (The independence-baseline excess is reliability times difficulty variance)**.** *With I the mean of $p_i p_j$ over ordered pairs,*

$$O - I = \frac{N\,\mathrm{Var}_m(f_m) + \mathrm{Var}_i(p_i) - \bar{f}(1-\bar{f})}{N-1} \;=\; \alpha_N \cdot \mathrm{Var}_m(f_m),$$

where α_N is Cronbach's α computed with the N models in the item role [Kuder & Richardson, 1937, Cronbach, 1951].

We verified the second equality numerically: residual $\leq 6.9 \times 10^{-17}$ across five shapes with $N \in [25, 1200]$, $M \in [45, 900]$, and -4.2×10^{-17} on the real ARC matrix, where $\alpha_N = 0.99928$. Since $\alpha_N \to 1$ for large N, the reported "excess co-failure" is, to an excellent approximation, simply the cross-item variance in difficulty — strictly positive whenever items differ in difficulty, including when every model fails independently. The asymptotic statement is the de Finetti / beta-binomial identity, and Yule [1903] is the classical warning about association induced by mixing.

For multi-category responses the analogous statistic is Fleiss's observed agreement [Fleiss, 1971], the rate at which two models give the same wrong answer given both are wrong — the statistic Kim et al. [2025] compare against a uniform $1/(K-1)$ baseline.

Lemma 3 (Agreement-given-both-wrong depends only on per-item response composition)**.** *Let n_{mk} be the number of models choosing option k on item m and $n_m = \sum_k n_{mk}$ the number of models observed on that item. Then the mean pairwise same-answer rate is*

$$A \;=\; \frac{\sum_m \sum_k n_{mk}(n_{mk}-1)}{\sum_m n_m(n_m-1)},$$

a function of the per-item response composition $\{n_{mk}\}_k$ alone; consequently any randomisation that preserves each item's response composition — permuting which model gave which recorded answer, item by item — leaves A exactly invariant, with zero variance.

The proof is the multi-category analogue of Lemma 1: A is a sum-of-squares statistic over the $\{n_{mk}\}$ counts alone, so it is a deterministic function of those counts and invariant under any relabelling that preserves them. We verify this constructively rather than restating it abstractly: in a simulation where models err *conditionally independently* given the item, drawing from a shared per-item distractor-attractiveness distribution, restricted to the subset where both models are wrong we observe agreement 0.5567 against a uniform baseline of 0.3333 — an apparent excess of +0.2233 — while the excess over the composition-preserving null implied by Lemma 3 is -1.1×10^{-16}. This is the founding premise of answer-copying detection [Wollack, 1997, van der Linden & Sotaridona, 2006], and we restate it rather than claim it.

Why restate known results. Because the model-evaluation literature cites none of them, and because the three statements are the same theorem in three notations. Making the bridge explicit is the cheapest available correction to a measurement practice now in active use.

4 The canonical null

Jo et al. [2026] prove that any distribution on $\{0,1\}^m$ is a mixture of conditionally independent Bernoullis, so excess dependence shrinks monotonically as the null grows more expressive; they conclude the choice is subjective. We observe that the ladder has a distinguished rung.

The Rasch model $\Pr(F_{im} = 1) = \sigma(\alpha_i + \beta_m)$ has $\{r_i\}$ and $\{c_m\}$ as jointly sufficient statistics. Conditioning on both therefore eliminates α and β exactly, and the conditional law of F given its margins is *uniform* on the set of margin-matched binary matrices, for every parameter value [Rasch, 1960, Andersen, 1973]. That set is precisely the ecologists' fixed-fixed null, sampled here with curveball [Strona et al., 2014]; the equivalence between the two traditions is standard [Besag & Clifford, 1989, Ponocny, 2001, Verhelst, 2008, Miller & Harrison, 2013].

The practical consequence is that this null estimates nothing. A fitted IRT null requires a parameterisation, an optimiser, penalties, and an identification convention, each of which is a defensible-but-arbitrary analyst choice of exactly the kind Jo et al. [2026] identify. The conditional null has none. It is the strongest ability-and-difficulty control that can be applied without fitting, and it is unique in that respect.

Margin-conditioned excess. For a point estimate of the null mean we use the maximum-entropy (Rasch) mean field $P_{im} = \sigma(\alpha_i + \beta_m)$, fitted to a maximum absolute margin error of 5.7×10^{-13}, and

$$D \;=\; \frac{FF^\top - PP^\top}{M}.$$

The mean field is validated against the randomisation itself: over 319,600 model pairs the curveball expectation of C_{ij} and the Rasch prediction correlate at 0.99999, mean absolute difference 0.00039 on a level of 0.2849. The matrix D is the excess matrix of the network-backbone literature [Neal et al., 2021].

Definition 1 (Effective number of independent models)**.** Let R be D scaled to unit diagonal, with eigenvalues λ_k. Then $N_{\text{eff}} = (\sum_k \lambda_k^+)^2 / \sum_k (\lambda_k^+)^2$.

The participation ratio, and the practice of removing a dominant shared mode before reading the bulk, are standard in econophysics [Laloux et al., 1999, Plerou et al., 1999]; the same quantity is M_{eff} in statistical genetics [Cheverud, 2001, Nyholt, 2004, Li & Ji, 2005]; eigen-decomposing Rasch residuals is standard psychometrics [Linacre, 1998]. Our contribution here is the calibration — N_{eff} must be compared against exact-margin replicates, because its null value is emphatically not N — and the application to model redundancy.

5 Data

We harvest per-item outcomes from the archived Open LLM Leaderboard. Only the metric column of each file is read — 1.4 KB out of a 63 MB file for HellaSwag — which makes an ecosystem-scale harvest possible at zero inference cost.

Two pitfalls had to be handled and are worth recording. Item identifiers are not stable: the harness stored a dataset identifier in mid-2023 and the raw question text thereafter, so a naive identifier join returns an *empty* intersection. The accuracy field also moved into a nested `metrics` struct in the 2024 schema. We resolve item identity by row index, licensed by explicit verification — across all five benchmarks, row order was identical for every one of 334 models sampled evenly over July 2023 to May 2024, with zero mismatches (ARC 149/149, TruthfulQA 60/60, GSM8K 60/60, Winogrande 40/40, HellaSwag 25/25) — and enforce a row-count guard on every read. Two limits are worth stating: this is a sample of $\approx$ 1,400 models, and only the ARC and HellaSwag samples span both identity-column schema generations, so those two are what actually exercise the drift described above. An undetected misalignment would make a model's row look independent of every other and so bias correlations *toward* the null, understating rather than manufacturing our result.

Benchmark	Models	Items	Mean accuracy	Snapshots
ARC-Challenge	1362	1165	0.527	2023-07 – 2024-06
Winogrande	1361	1267	0.734	2023-09 – 2024-06
TruthfulQA (mc1)	1334	786	0.368	2023-07 – 2024-06
GSM8K	1228	1319	0.348	2023-09 – 2024-06
HellaSwag	1362	9404	0.581	2023-07 – 2024-06

Table 1: Model and item counts after removing degenerate rows and columns; mean accuracy is over the full pre-filter matrix. For comparison, Jo et al. [2026] analyse 451 models, Kim et al. [2025] 349.

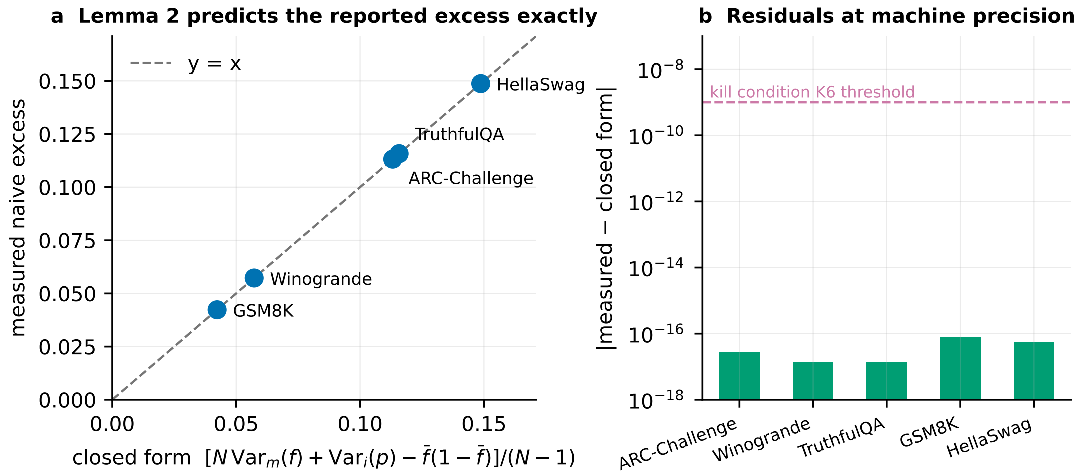

Figure 1: Lemma 2 predicts the measured excess exactly on all five benchmarks (panel a); the residual sits at machine precision, well below the 10^{-9} validation threshold set in the pre-registration (panel b).

6 Results

6.1 The closed form is exact on real data

Measured excess versus Lemma 2: ARC +0.113180719 vs +0.113180719 (residual -2.8×10^{-17}); Winogrande +0.057265401 (-1.4×10^{-17}); TruthfulQA +0.115767366 ($+1.4 \times 10^{-17}$); GSM8K +0.042311309 ($+7.6 \times 10^{-17}$); HellaSwag +0.148717716 (-5.6×10^{-17}). Over 400 curveball replicates per benchmark, with both margins exactly preserved, the maximum deviation of mean co-failure from its observed value is 0.00×10^{0}, as Lemma 1 requires. Of a reported excess as large as +0.116, none survives conditioning.

6.2 The pre-registered dispersion statistic is refuted

Our pre-registered primary test was whether $T = \mathrm{Var}_{i<j}(C_{ij})$ exceeds its null. It does not do so consistently (Table 2); the pre-registered kill condition required the effect in at least two of three primary benchmarks and it fired. We report T as uninformative rather than reinterpreting it.

6.3 What survives: effective independence

Conditioning on the margins removes everything Lemma 1 and Lemma 2 describe, yet the residual spectrum of D is far from flat. Table 3 reports the participation ratio of that spectrum against its exact-margin null on all five benchmarks: the observed value sits at 2.8–21.4% of

Benchmark	T observed	T null	SES	ratio
ARC-Challenge	7.640×10^{-3}	8.331×10^{-3}	-70.9	0.917
Winogrande	1.310×10^{-3}	2.322×10^{-3}	-149.8	0.564
TruthfulQA	1.277×10^{-2}	1.164×10^{-2}	$+80.1$	1.098
GSM8K	4.576×10^{-2}	4.623×10^{-2}	-38.5	0.990
HellaSwag	5.098×10^{-3}	5.085×10^{-3}	$+10.6$	1.002

Table 2: Independent-chain nulls, $R = 200$ chains each burned in 50 trades per model.

Benchmark	N	N_{eff} obs.	N_{eff} null	ratio	raw (unconditioned)
ARC-Challenge	1362	23.6	449.3 ± 3.4	0.052	1.9
Winogrande	1361	20.1	493.8 ± 3.0	0.041	3.6
TruthfulQA	1334	17.4	309.7 ± 2.4	0.056	1.5
GSM8K	1228	118.6	553.1 ± 2.2	0.214	2.1
HellaSwag	1362	27.3	958.0 ± 3.2	0.028	1.5

Table 3: The last column shows why calibration is not optional: uncalibrated, the participation ratio reads 1.5–3.6, which is not a finding but the single shared difficulty eigenvalue.

the null, a gap of the same order everywhere despite the benchmarks spanning a 12× range in item count. The last column shows why this comparison must be calibrated rather than read off directly — the raw, unconditioned participation ratio is 1.5–3.6 on every benchmark, a number driven almost entirely by the single shared item-difficulty eigenvalue that conditioning is designed to remove. Only once that eigenvalue is subtracted out does the residual gap become visible at all. What that gap means — and what it does not license claiming — is the subject of the rest of this section.

6.4 Why these numbers are not an "effective number of models"

It is tempting to read Table 3 as "1,362 models behave like 24". That reading is wrong, and the reason is algebraic. For any unit-diagonal R,

$$\mathrm{PR} \;=\; \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} \;=\; \frac{N}{1 + (N-1)\,\overline{R_{ij}^2}},$$

which we confirm to machine precision on ARC (16.0449 computed either way). The eigendecomposition contributes nothing: the participation ratio is a monotone transform of the mean squared off-diagonal residual correlation. An equicorrelated matrix — one weak global factor plus $N-1$ independent directions — reproduces the reported value exactly, and so does a partition into 24 blocks of exact clones. The statistic cannot tell them apart, so it cannot license a count of independent models.

The diagnostics that can tell them apart give a clear answer on ARC: $\lambda_1^2 / \sum \lambda_k^2 = 0.540$ (one-factor ≈ 0.99, clone blocks ≈ 0.10); deflating the leading eigenvector raises PR only from 16.0 to 46.8, not to $\approx N$; and six observed eigenvalues exceed the null's maximum eigenvalue of 27.9. We therefore report the primitive quantity, $\mathrm{rms}\,|R_{ij}| = 0.2483$ against 0.0449 under the null, and the count of dimensions above the null spectral edge. The null value is the pooled mean of four dispersed curveball chains (25 draws each, $\hat{R} = 0.988$; Section 6.7), and agrees with three independent estimates of it to within 0.0004.

Independent 2026 work reaches the same caution about this exact statistic from a different direction. Sha & Zhao [2026] apply the participation ratio of a benchmark-score spectrum to count effective independent *benchmarks* within an evaluation suite (BBH vs. MMLU-Pro, $\rho = 0.96$) and explicitly warn that "binary spectra overestimate absolute latent dimensionality,"

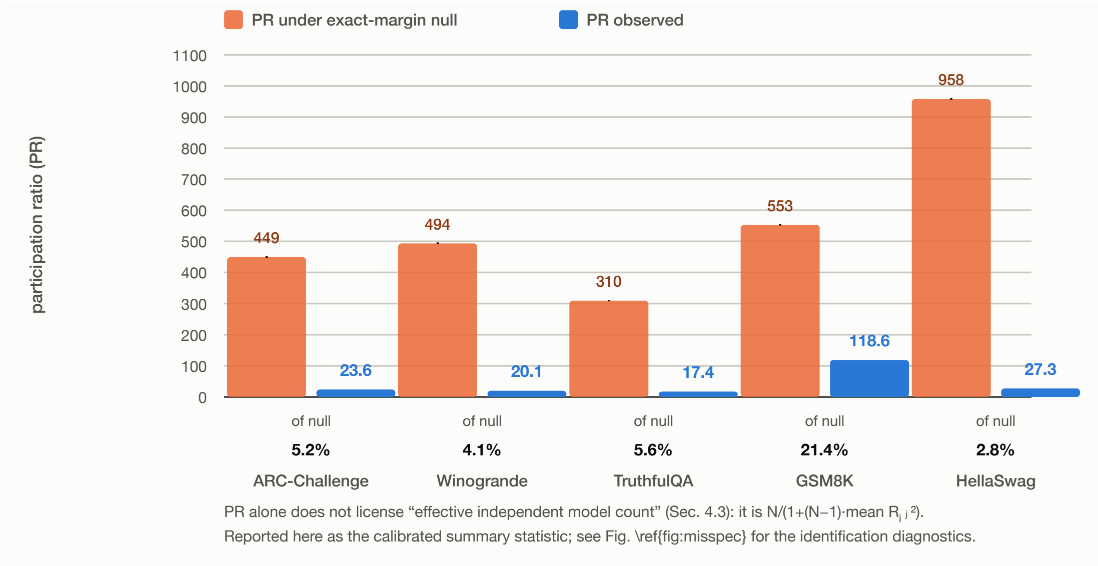

Figure 2: Observed participation ratio against its exact-margin null, all five benchmarks. Read this alongside Section 6.4: PR is reported here as a calibrated summary statistic, not as a count of independent models.

Generating process	PR / $\mathrm{PR}_{\mathrm{null}}$	rms $\lvert R_{ij}\rvert$	$\lambda_1^2/\sum\lambda^2$	dims > edge
1PL, correctly specified	1.001	0.065	0.105	1
2PL, $\mathrm{sd}(\log a) = 0.35$	0.966	0.068	0.119	1
2PL, $\mathrm{sd}(\log a) = 0.60$	0.862	0.076	0.167	1
two ability dimensions	0.230	0.162	0.457	3
20 exact clone clusters	0.112	0.234	0.102	19
real ARC	**0.052**	**0.248**	**0.540**	**6**

Table 4: Discrimination heterogeneity alone cannot produce the observed effect. A low-dimensional multi-ability structure produces much of it, and the observed spectral signature resembles that case rather than duplicated model families.

treating their statistic as a screening diagnostic rather than a literal factor count — the same epistemic move we make here, on the orthogonal axis (benchmarks rather than models) and without an exact conditional null.

6.5 Misspecification controls: what else could produce this

Our conditioning model is one-parameter. Any two-parameter or multidimensional structure leaves residual signal that the margin-preserving null cannot reproduce, so we ask how much of the effect that alone explains. Each row below is a simulated population with *no* clusters, no clones and no shared lineage — errors conditionally independent given the item.

Two conclusions follow. Discrimination heterogeneity is *not* a sufficient explanation: even at $\mathrm{sd}(\log a) = 0.60$ it yields rms 0.076 against the observed 0.248. But a small number of extra ability dimensions is a substantial one, and the observed $\lambda_1^2/\sum\lambda^2 = 0.540$ sits near the two-dimensional case (0.457) and far from the clone case (0.102). We therefore state the finding as: after conditioning on item difficulty and unidimensional ability, open-model failure retains correlated structure of roughly six dimensions and a 5.5-fold excess rms correlation on ARC (2.8–11.8× across the five benchmarks), whose signature is that of additional latent ability dimensions rather than of duplicated models.

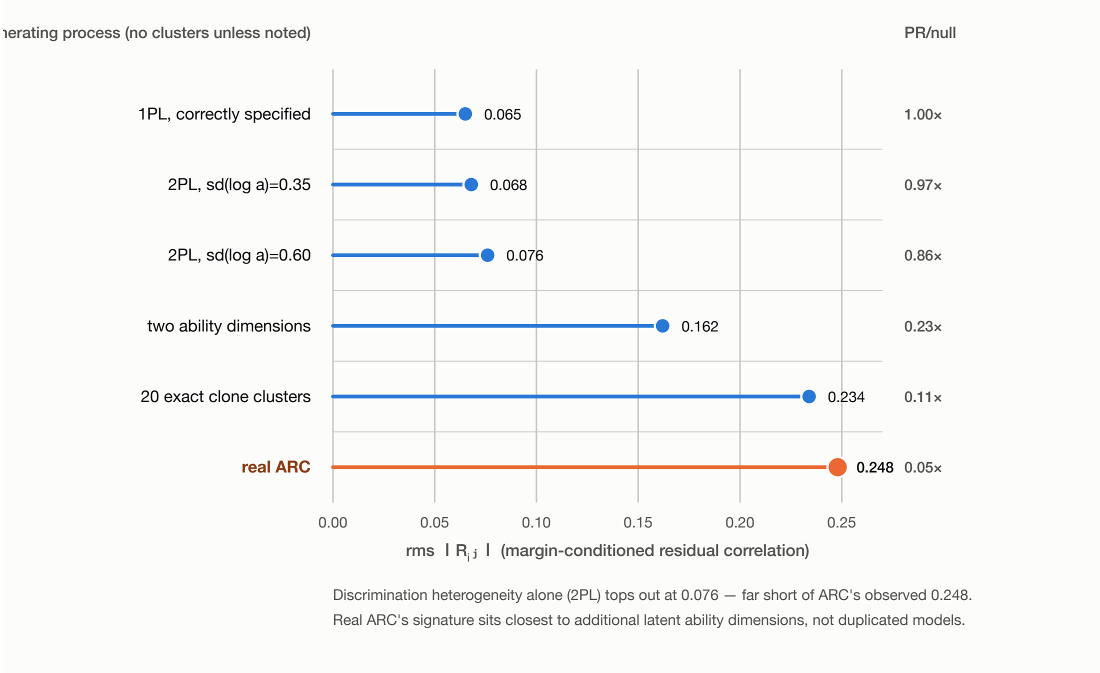

Figure 3: Identification evidence: real ARC's residual correlation (orange) sits far beyond what two-parameter discrimination heterogeneity alone can produce, and its $\lambda_1^2/\sum\lambda^2$ signature (Table 4) matches a multi-ability structure rather than duplicated models.

6.6 Reconciling two opposite-looking results

T below null and N_{eff} far below null appear to conflict: since R has unit diagonal, $\operatorname{tr} R = N$ exactly and $N_{\text{eff}} = N^2/\|R\|_F^2$, so N_{eff} far below null is algebraically equivalent to the off-diagonal excess correlations being far *above* null. The resolution is exact. Because the null preserves margins, the fitted P — and hence $E = PP^\top/M$ — is identical for the observation and every replicate; we measure $|\operatorname{Var}(E)_{\text{obs}} - \operatorname{Var}(E)_{\text{null}}| = 0.000 \times 10^0$. Writing $C = E + D$,

$$\operatorname{Var}(C) = \operatorname{Var}(E) + 2\operatorname{Cov}(E, D) + \operatorname{Var}(D).$$

On ARC: $\operatorname{Var}(D) = 6.253 \times 10^{-4}$ against a null of 2.127×10^{-5} (SES $+1613$, a $29\times$ excess), while $\operatorname{Cov}(E, D) = -6.494 \times 10^{-4}$ against -2.1×10^{-6} (SES -153; correlation -0.285 versus -0.005). The negative covariance outweighs the excess variance, which is exactly why T reads below null. Both statistics are correct and the arithmetic closes to the reported $\operatorname{Var}(C)$.

Substantively: pairs that the margin model predicts will co-fail most actually co-fail relatively less, and vice versa. That is structured deviation from unidimensionality — which is what "models cluster into a few dozen behavioural types" means when stated in Rasch terms. We flag that "residual structure" and "Rasch misfit" are the same observation under two descriptions, and that our design cannot separate them.

6.7 Robustness

Near-duplicates. The archive contains exact duplicates (maximum pairwise agreement 1.0000). Clustering models by agreement and keeping one representative per cluster: on ARC, removing 900 of 1,362 models moves N_{eff} from 23.6 only to 29.0; on TruthfulQA, removing 1,001 of 1,334 moves it from 17.4 to 26.2; on GSM8K, removing 450 moves it from 118.6 to 111.4. Redundant submissions are not the mechanism.

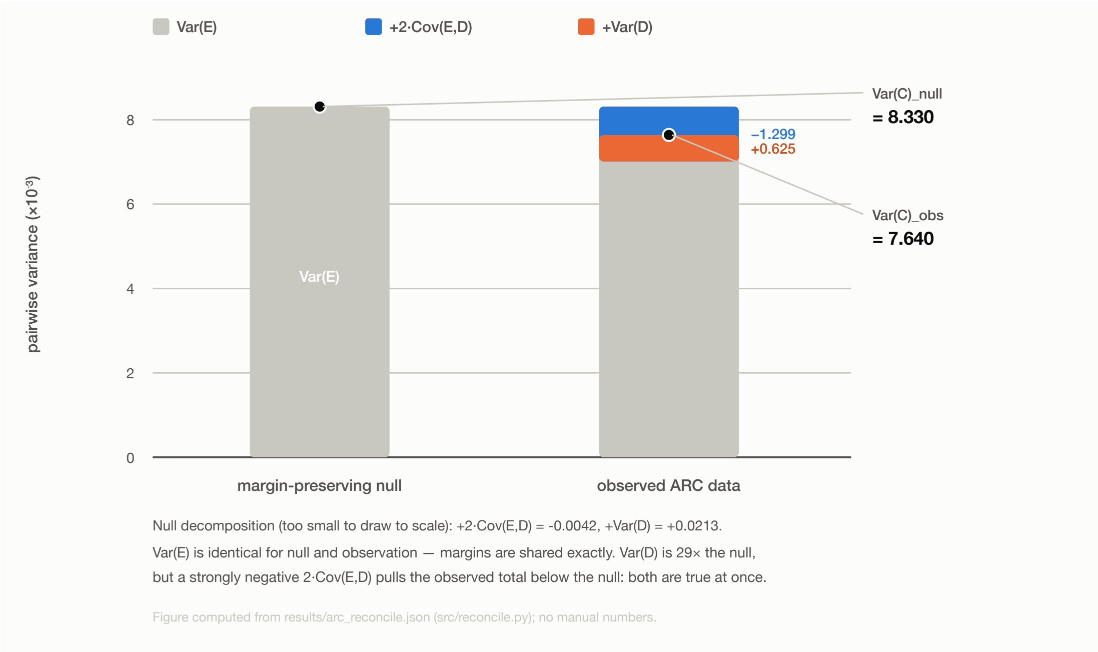

Figure 4: The exact decomposition $\mathrm{Var}(C) = \mathrm{Var}(E) + 2\,\mathrm{Cov}(E, D) + \mathrm{Var}(D)$ on ARC. $\mathrm{Var}(E)$ is identical for null and observation by construction; the observed column's negative covariance term outweighs its 29×-inflated variance term, landing below the null.

Mixing. T plateaus by about 10 trades per model and is flat to 200. Independent chains give null SD ratios 0.96–1.02 versus single-chain thinning, with lag-1 autocorrelations between -0.14 and $+0.07$. A 2000-trades-per-model burn-in gives an ARC N_{eff} null of 450.8 against 449.3 at 50 — a 40× longer burn-in moves it by 0.3%. Both margins are exactly preserved at every checkpoint.

The sampler is uniform, verified rather than asserted. Carstens [2015] proved curveball converges to the uniform distribution on the fibre provided failed trades are counted as steps; our implementation does so. We test it directly. Enumerating every binary matrix with given margins on four small fibres (sizes 48, 90, 204, 1170) and drawing 60,000 thinned samples, a χ^2 test against uniformity is not rejected on any of them ($p = 0.05, 0.23, 0.73, 0.15$). Across all five benchmarks, four dispersed chains give Gelman–Rubin $\hat{R}$ between 0.98 and 1.03 at the burn-in actually used.

The null is not the finite-sample noise floor. A natural objection is that the exact null's residual-correlation rms simply reproduces the $1/\sqrt{M}$ sampling floor. It does not, but the margin is modest: the exact null sits at 1.25–2.01× the analytic floor (0.0446 vs 0.0293 on ARC; 0.0207 vs 0.0103 on HellaSwag). Two further rungs are informative. Conditioning on model margins alone lands *exactly* on that floor on every benchmark, while conditioning on item margins alone reproduces the full both-margin null to within 8.3% on every benchmark, and to within 0.4% on ARC (0.0445 vs 0.0446); the largest gap is HellaSwag's. Empirically, the item margins do essentially all of the conditioning work; the joint-sufficiency framing is correct as stated but overstates what is load-bearing here.

Deduplication does not drive the result. Recomputing every headline statistic on populations deduplicated at agreement thresholds 0.99 through 0.90, refitting the Rasch model and redrawing the null each time, 14 of 15 statistic–benchmark pairs move by less than 1.5× between

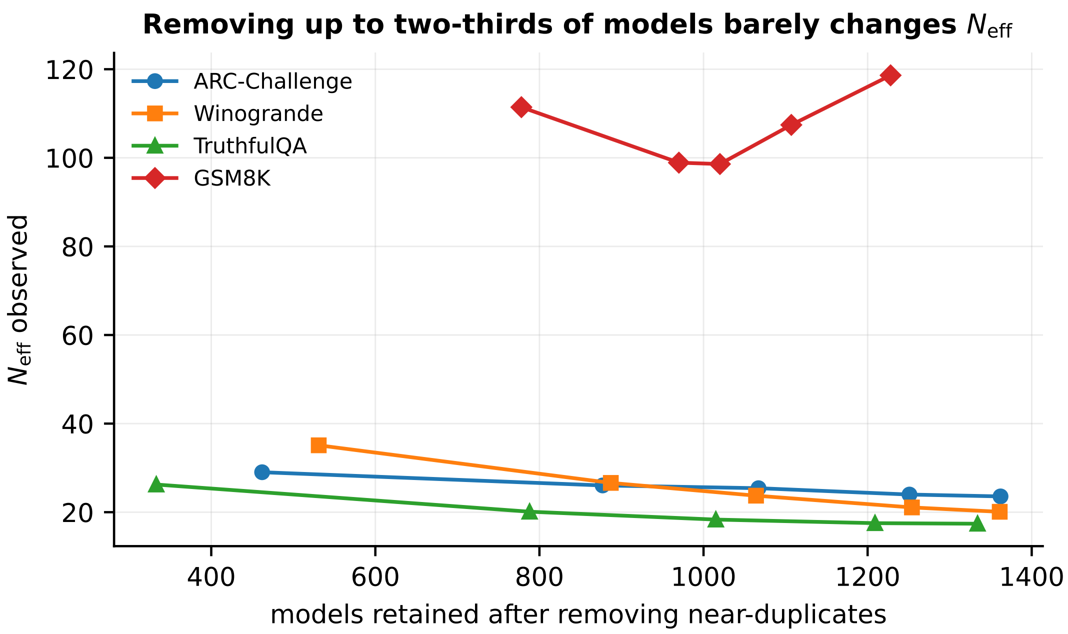

Figure 5: Observed participation ratio as near-duplicate models (agreement $\geq$ threshold, one representative kept per cluster) are progressively removed. Removing up to two-thirds of each benchmark's models moves the statistic by only a few units.

the full population and the 0.95-deduplicated one, which discards 36–64% of models. The exception is HellaSwag's participation ratio (2.44×). At that same 0.95 threshold, the eigenvalue count above the null edge matches the full-population count on three of five benchmarks (ARC, Winogrande, GSM8K); it is off by one on TruthfulQA and by four on HellaSwag.

How many dimensions, and by a method that needs no fitting. Counting eigenvalues above the null spectral edge is Horn's parallel analysis with a stronger reference distribution than Horn's own: instead of comparing against iid random data, we compare against the exact margin-preserving null, which already carries the observed ability and difficulty profiles. It requires no optimiser and inherits the sampler guarantees above. The retained counts are 6 (ARC), 5 (Winogrande), 5 (TruthfulQA), 4 (GSM8K) and 12 (HellaSwag), and they are broadly stable under deduplication — across thresholds 0.99 to 0.90 they move by at most one on three of the five benchmarks (Winogrande, TruthfulQA, GSM8K), while ARC ranges 4–6 and HellaSwag ranges 8–12. We report this as the dimensionality estimate and note its known limitation: eigenvalue-counting rules of this family tend to over-retain, so these are upper bounds rather than point estimates. We attempted a complementary estimate by fitting multidimensional IRT and comparing held-out synthetic residual correlation; three parameterisations disagreed materially and the attempt is reported in the repository as unresolved rather than folded in here.

Power, and one blind spot. On planted data (400 models, 700 items, 20 datasets per condition, $\alpha = 0.05$; null conditions reject at 0.017), the test reaches power 1.00 against shared failure modes at a planted rms of 0.048 — far below the 0.099–0.294 observed — and detects copying of a reference model with power 0.65 at a copy rate of 0.05 and 1.00 from 0.10. Against the alternative in which all models share one item-difficulty profile and differ only in ability, power is 0.05: exactly α, because that alternative is the null (Figure 6). We state this as a property of the design rather than a caveat [Colwell & Winkler, 1984].

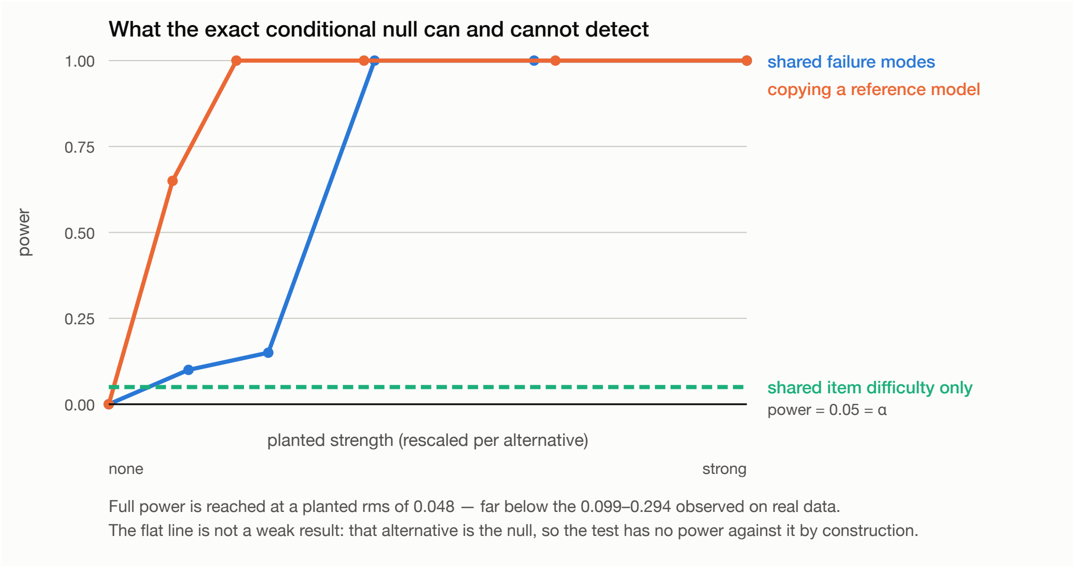

Figure 6: Detection power against three planted alternatives (`results/power_study.json`). The dashed line is the design's blind spot rather than a weak result: a population in which every model shares one item-difficulty profile and differs only in overall ability is exactly what the null describes, so no test conditioning on the item margins can separate it from chance.

Functional choice. On ARC, 91.35% of spectral mass is positive. The participation ratio is insensitive to the treatment of negative eigenvalues: 23.6 clipped, 23.5 absolute, 16.0 raw.

Estimator controls. On a matrix drawn from a Rasch margin model the dispersion estimator reports SES +0.05; on eight latent model families +6.75; on exact clones +2.64; on column-shuffled real data −1.33.

6.8 Real responses: the accuracy–agreement claim survives conditioning

Everything above concerns *correctness* (binary failure). Lemma 3 (Section 3) concerns *which wrong answer* two models give, and predicted — and, in simulation, demonstrated — that this can look like "more correlated errors" purely from per-item distractor attractiveness, with no shared behaviour at all. We pre-registered a direct test of whether that critique undermines Kim et al. [2025]'s specific empirical claim, with a kill condition (K7) stated in advance: if the accuracy–agreement slope survives margin conditioning with confidence interval excluding zero, our critique does not apply here, and we report their finding as robust. We reconstruct each model's chosen answer from per-choice log-likelihoods on a 600-model ARC subsample, recovering the gold answer from cross-model agreement where models are correct (agreement among correct models 1.0000 on 1,144/1,172 items; the recovered gold reproduces each model's own reported accuracy on 99.01% of cells, which we take as validating the reconstruction rather than assuming it).

K7 fires. The raw slope of agreement-when-both-wrong against mean pair accuracy is +0.585 (95% CI $[0.577, 0.594]$); conditioned on the composition-preserving null it is +0.514 ($[0.506, 0.522]$) — an attenuation of only 12.1%. The claim that more accurate models give more similar wrong answers is not an artifact of item-level distractor composition: it survives our conditioning almost fully. We report this as prominently as the alternative, per the pre-registration.

Two things are true at once, and both matter. The mean-level excess vanishes: 0.7332 observed versus 0.7324 expected under conditional independence (+0.00085), essentially zero, echoing Lemma 3's degeneracy result. But the *relationship between accuracy and agreement* is not a mean-level statistic, and it is not absorbed by the same conditioning. Composition-based conditioning and accuracy-slope conditioning are different operations; only the latter bears on Kim et al.'s claim, and it does not remove it.

7 Related work

Kim et al. [2025] measure agreement-when-both-wrong across 349 models against a uniform baseline, conditioning on model accuracy but not item difficulty, and report that larger and more accurate models have more correlated errors. Jo et al. [2026] show that conditioning on item difficulty via fitted IRT attenuates such correlations substantially and sometimes reverses their sign, and argue the inference is null-dependent and hence subjective; our contribution is to identify a null on that ladder which requires no fitting, and to report the answer there.

Closest in spirit is Kohli [2026], who quantify the effective independence of a 9-judge, 7-family LLM-as-judge panel on ChaosNLI and RewardBench and find roughly two effective votes. The relationship to our work is worth stating precisely, because the questions coincide and the machinery does not. Their nulls are a difficulty-stratified permutation that preserves per-judge error rates within entropy bins, and a Condorcet null simulated from fitted per-judge, per-bin confusion matrices; both estimate a nuisance model, which is exactly the analyst choice Section 4 is designed to remove, and neither preserves the item margins exactly. Their headline statistic is the Kish design effect $n_{\text{eff}} = k/(1 + (k-1)\bar{\phi})$. That is the same algebraic form as the participation-ratio identity of Section 6.4, differing in that Kish uses the mean pairwise correlation where the participation ratio uses the mean *squared* correlation. Our negative result therefore transfers: a statistic of this form is a monotone function of a single summary of the correlation matrix and cannot separate one weak global factor from many tight clusters. They note the exchangeability assumption this rests on and check it against a largest-eigenvalue variant; we would add that the check does not rescue the interpretation, only the arithmetic. The two studies are complementary in scale as well: 9 judges with 100 human annotations per item against our 1,228–1,373 models with no human labels.

Where this bears on evaluation practice. A growing line of work fits item-response models to model-by-item outcome matrices in order to compress benchmarks: Polo et al. [2024] estimate full-benchmark performance from roughly 100 items using IRT models trained on the evaluation results of 319 models, and subsequent work extends the idea to anchor-item calibration and adaptive testing. These methods are the most direct consumers of the object we study, and they rest on an assumed latent dimensionality. Our estimate — 4 to 6 dimensions above the exact null's spectral edge on four benchmarks, and 12 on HellaSwag — is therefore not only a statement about monoculture; it is a statement about whether a low-dimensional ability parameterisation is adequate for the population these methods are calibrated on, over a model set roughly four times larger than that used to fit them. The test we give requires no IRT fit of its own, so it can be used to check that assumption rather than presupposing it.

Foundational framing of algorithmic monoculture and outcome homogenisation is due to Kleinberg & Raghavan [2021] and Bommasani et al. [2022]. On method, the fixed-fixed null and curveball come from ecology [Connor & Simberloff, 1979, Schluter, 1984, Gotelli, 2000, Strona et al., 2014]; the equivalent Rasch conditional machinery from psychometrics [Rasch, 1960, Andersen, 1973, Besag & Clifford, 1989, Ponocny, 2001, Verhelst, 2008, Miller & Harrison, 2013]; the participation-ratio and shared-mode-removal apparatus from econophysics [Laloux et al., 1999, Plerou et al., 1999] and statistical genetics [Cheverud, 2001, Nyholt, 2004, Li & Ji, 2005]; and the excess-matrix construction from network backbone extraction [Neal et al., 2021].

8 Limitations

Claims cover the evaluated open-model ecosystem on multiple-choice-style benchmarks, not deployed proprietary systems; the societal framing is motivation, not a measured outcome, and we make no claim about real-world screening decisions. The models are a self-selected population of leaderboard submissions, and Jo et al. [2026] show explicitly that such inferences are population-dependent — a different peer set can change them.

The central limitation is identification. Our conditioning model is unidimensional, and Table 4 shows that additional latent ability dimensions reproduce much of the observed effect with no shared lineage whatsoever. We can rule out discrimination heterogeneity as a sufficient explanation, and we can rule out duplicate models, but we cannot separate "genuine behavioural concentration" from "the Rasch family is too small" — and on the spectral evidence the latter is the better description. Re-conditioning on a fitted 2PL or low-rank logistic model would sharpen this at the cost of the exactness that motivates the design; that trade-off is the natural next step.

We also report a negative methodological result about our own pre-registered plan: the participation ratio does not measure an effective number of independent models (Section 6.4), and we have removed that interpretation rather than retain it.

Pre-registered kill condition K5 failed — declared `base_model` coverage was 23.3% against a 40% threshold, with 315 of 1,362 model repositories no longer retrievable — so the lineage decomposition is dropped rather than estimated on a biased subset.

Section 6.8's real-response result is a limitation on the paper's own critique, stated plainly: Lemma 3 shows composition alone *can* manufacture apparent agreement, but on real ARC responses the specific published claim it was aimed at does not turn out to be an instance of that artifact. The two findings do not contradict — they concern different statistics, mean-level agreement versus the accuracy–agreement slope — but a reader taking only the abstract's mention of this degeneracy could reasonably expect more debunking than the data supports, which is why the full result is reported in Section 6.8 rather than folded silently into the mean-level finding.

9 Reproducibility and use of AI assistance

All code, the harmonised substrate, and every results artifact are released with a datasheet (Gebru et al. schema) and Croissant 1.0 metadata; each number traces to a JSON file emitted by an executed run, and the propositions are asserted as unit tests. The study was pre-registered before the confirmatory analysis, with dated amendments recorded before any confirmatory statistic was computed; one pre-registered hypothesis was refuted by its own kill condition and is reported as refuted. Every kill condition in the later robustness work was likewise written before its experiment ran, and four of them fired — including one that retired an estimate of latent dimensionality by multidimensional IRT after three parameterisations disagreed, which is why Section 6.7 reports dimensionality by parallel analysis instead.

The upstream license chain is documented rather than assumed: ARC-Challenge is CC-BY-SA-4.0 and therefore share-alike, TruthfulQA is Apache-2.0 and GSM8K is MIT, while HellaSwag, Winogrande and the leaderboard archive itself declare no license. The reconstructed ARC answer key is withheld from the public release — it is a derivative of a share-alike work and republishing a benchmark answer key is a contamination vector — and the recovery code is released in its place.

This work used a large language model (Claude, Anthropic) as a coding and analysis assistant: it wrote the harvesting, estimation and figure code and assisted in drafting and in the adversarial prior-art review that led to Section 3 being rewritten from claimed novelty to attribution. The author directed the research, specified and pre-registered the hypotheses and kill conditions, verified the derivations, and takes full responsibility for the content, in line with COPE guidance that AI tools cannot be authors.

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